Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Additives derived from lignin for recyclates extending their service life			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Chemicals		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	Polymeric Materials	Polymer Recycling Technologies	Fine Chemicals	Additives
Concept of Proposed Technology	The proposed technology focusses on bio-based stabilizers for recycled polyolefin plastics. Building on Fraunhofer CBP's long-standing expertise in lignin fractionation and chemical modification [1,2] and Fraunhofer LBF's track record in polymer recycling and bio-based stabilizer systems [3,4], tailored lignin derivatives are developed that can be used as high-performance additives protecting recycled plastics during processing and extending their service life. This enables manufacturers to increase the share of recycled material in products without sacrificing quality, color, or mechanical properties. The technology can be applied in a wide range of plastic products, including packaging, consumer goods, automotive parts, construction materials, and components for renewable energy			

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	systems.
Rationale for Joint R&D Project	The development and market introduction of lignin-based stabilizers for recycled plastics require the combination of advanced materials research with strong industrial implementation and market access. Fraunhofer contributes deep expertise in lignin chemistry, process development, and performance evaluation of recycled polymers, while Korean SME partners provide application know-how, compounding and processing capabilities, and direct links to regional and global plastic markets. Only through this joint R&D project can the technology be adapted to industrially relevant formulations, scaled to production conditions, and validated in real customer applications. This collaboration significantly shortens time-to-market, reduces technical risk, and ensures that the resulting solutions fit both European and Asian regulatory and market requirements.
Key Words	1. Lignin modification 2. Thermal stability, antioxidants 3. Enhancing sustainability of plastic recycling processes 4. Enhancing the performance of recycled plastics 5. bio-based additives

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3-4)	End of the Project	Level (7-8)
R&D Objectives	The R&D objectives are to design and optimize chemically modified lignin additives that deliver reliable processing and long-term thermal stabilization in recycled polyolefin compounds. The partners will establish scalable synthesis and formulation routes, moving from lab-scale chemistry to pilot-scale production that is compatible with industrial compounding and extrusion lines. A further objective is to integrate these lignin-based stabilizers into application-relevant masterbatches and compounds and to benchmark their performance against state-of-the-art commercial stabilizer systems. Finally, the project aims to generate validated data packages and prototype products for key markets such as packaging, automotive, consumer goods, and construction, creating a solid technical foundation for rapid commercialization by Korean SME partners.			
R&D Target Specifications	The R&D project targets lignin-based stabilizers to replace conventional fossil-based stabilizers in recycled polyolefin compounds while maintaining equivalent or better performance. At typical additive loadings below 1 wt%, the materials should achieve an oxidation induction time (OIT) of more than 20 minutes at 200 °C and retain at least 85% of their initial tensile strength after 500 hours of accelerated thermal aging at 80 °C. The compounds are expected to withstand at least 3–5 extrusion or injection molding cycles with less than 10% loss in mechanical properties and a color change ΔE below 3. In addition, the technology should enable the use of more than 50% recycled content in selected applications without falling below current industrial quality and processing standards.			
R&D Plan	The R&D plan spans 36 months and is divided into a six-month Phase 1 and a thirty-month Phase 2. In Phase 1, **<u>Phase 1</u>** **Work Package 1** (Fraunhofer) focuses on preliminary trials at the 100 g scale, using recycled polyolefins supplied by the Korean partners in parallel with comparable recyclates sourced from Europe. These materials are compounded with selected lignin-based stabilizers and evaluated for processing behavior and basic stabilization effects, providing a first mapping of structure–property relationships. **Work Package 2** (Korean SME). The Korean SMEs characterize the quality, variability, and contaminant profiles of their own recyclate streams using industrially relevant methods, ensuring that the subsequent development work is aligned with realistic feedstock conditions and priority applications. A key milestone at the end of Phase 1 is the joint selection of representative recyclates and initial lignin modification concepts that meet minimum performance thresholds. **<u>Phase 2</u>** Phase 2 extends over 30 months and is dedicated to systematic scale-up, validation, and preparation for industrial implementation. **Work Package 3** (Fraunhofer) The most promising lignin modification routes are transferred from lab conditions to a robust synthesis at the kilogram scale, optimizing reaction parameters, purification steps, and reproducibility.			

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Work Package 4 (Fraunhofer) scales up compounding to a pilot level, producing several tens of kilograms of stabilized recyclate on twin-screw extruders under conditions representative of industrial practice.

Work Package 5 (Fraunhofer). Standardized test specimens are produced from these compounds, followed by comprehensive mechanical characterization and long-term stability studies, including oxidation induction time, accelerated thermal aging, and color stability, to benchmark the new additives against state-of-the-art stabilizer systems.

Work Package 6 (Fraunhofer). The focus shifts to multiple reprocessing cycles, in which the stabilized recyclates are extruded or injection molded several times and then characterized again using the same mechanical and aging protocols as in AP5, in order to quantify performance under realistic circular economy scenarios.

Work Package 7 (Korean SME) Building on the results of WP6, the most successful compound formulations will be transferred to full production scale on the partners' industrial lines, fine-tuning processing parameters and producing larger batches of material that are converted into demonstrator parts for target applications such as packaging, consumer goods, or automotive components.

Work Package 8 (Korean partners, with support from Fraunhofer) carries out application-oriented testing of these parts under relevant use conditions, including mechanical performance, environmental stress, and customer-specific validation, and feeds the results back into a techno-economic and sustainability assessment.

The main milestones of Phase 2 are robust pilot-scale synthesis and compounding protocols, validated performance data sets, industrial demonstrator parts, and an implementation roadmap that enables rapid commercialization by the Korean SME partners after project completion.

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		Year 1				Year 2				Year 3			
Work package		Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
1,2	Evaluation												
3	Lignin derivatives												
4	Compounding of recyclates												
5	Testing												
6	Multiple recycling												
7	Industrial process												
8	Demonstrator and testing												
Milestones													
1	Providing of lignin derivatives												
2	Performance analysis												
3	Formulation & recipe development												

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institutes, represented by Fraunhofer CBP and Fraunhofer LBF, will lead the scientific and technological development of lignin-based stabilizers and their integration into recycled polyolefin compounds. The core responsibilities of CBP include the selection and characterization of suitable lignin feedstocks, the design and optimization of defined hydrolysis steps. This will be supported by LBF with chemical modification routes. Both Institutes will stepwise scale-up their processes from laboratory to kilogram scale. Fraunhofer LBF will develop and operate pilot-scale compounding processes, produce standardized test specimens, and carry out comprehensive mechanical, thermal, and aging tests to benchmark performance against conventional stabilizer systems. Both institutes will provide advanced analytical infrastructure, experienced scientific and technical staff, and established quality and safety procedures to ensure reliable and reproducible results. In addition, Fraunhofer will coordinate the R&D roadmap, manage work packages, and support the Korean SME partners with technical guidelines, data interpretation, and the adaptation of formulations to specific applications and processing conditions, thereby ensuring a strong scientific foundation for subsequent industrial implementation.
Expected R&R of Korean SME Partner	The Korean SME partners are expected to play a central role in transferring the developed technology into real industrial environments and market-ready products. They will provide representative recycled polyolefin streams, characterize their quality and variability under practical conditions, and define priority application fields and performance requirements from a customer perspective. Building on the formulations and processing know-how developed by Fraunhofer, the SMEs will scale up compounding to production scale on their own equipment, optimize processing parameters, and produce larger material batches and demonstrator parts such as packaging items, consumer goods, or components for automotive and construction. They will conduct application-oriented testing and validation, including processability, mechanical performance, long-term behavior, and compatibility with existing production workflows and quality standards. Furthermore, the Korean partners will commit personnel, production time, and market access, interact with local customers and regulators, and provide continuous feedback on technical, economic, and regulatory constraints. Through this close collaboration, they will help align the technology with Korean and international market needs and prepare the ground for rapid commercialization after project completion.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The joint R&D project is expected to deliver a validated technology platform for lignin-based stabilizers that can be reliably used in recycled polyolefin compounds at industrial scale. Technological outcomes include robust synthesis routes for modified lignin additives, scalable compounding recipes for different recyclate qualities, and thoroughly characterized materials that meet or exceed the performance of conventional fossil-based stabilizer systems. The project will generate comprehensive data on processing behavior, mechanical properties, long-term thermal stability, and multiple-recycling performance, providing a solid scientific and technical basis for market introduction. On this foundation, the Korean SME partners will be positioned to commercialize new lines of bio-based stabilizer masterbatches and high-recycled-content compounds tailored to applications such as packaging, consumer goods, automotive components, and construction products. By enabling higher recycled content without sacrificing quality or processing efficiency, the technology offers clear differentiation potential and access to fast-growing markets driven by sustainability regulations and brand-owner commitments. Economically, this reduces dependence on volatile petrochemical additives, creates value from an underused by-product of the pulp and paper industry, and opens up new business opportunities for SMEs in both the Korean and international plastics value chain. At a broader level, the project supports the transition toward a circular plastics economy, contributes to CO_2 reduction targets, and strengthens technological collaboration between European research institutes and Korean industry, paving the way for follow-up projects and long-term partnerships.

6. References

(1) J. J. Bernhardt, B. Rößiger, T. Hahn, D. Pufky-Heinrich, *Kinetic modeling of the continuous hydrothermal base catalyzed depolymerization of pine wood based kraft lignin in pilot scale,* Ind. Crop. Prod. 159 (2021) 113119, https://doi.org/10.1016/j.indcrop.2020.113119

(2) U. Junghans, J. J. Bernhardt, R. Wollnik, D. Triebert, G. Unkelbach, D. Pufky-Heinrich, *Valorization of Lignin via Oxidative Depolymerization with Hydrogen Peroxide: Towards Carboxyl-Rich Oligomeric Lignin Fragments,* Molecules 25 (2020) 2717, https://doi.org/10.3390/molecules25112717

(3) J. Mayer, E. Metzsch-Zilligen, R. Pfaendner, *Novel multifunctional antioxidants for polymers using eugenol as biogenic building block,* Polym. Deg. Stab. 200 (2022) 109954, https://doi.org/10.1016/j.polymdegradstab.2022.109954

(4) K. Markus , E. Metzsch-Zilligen , R. Pfaendner, *Radical scavenging and processing stability of novel biobased stabilizers in polypropylene, Polym. Deg. and Stab. 222 (2024) 110709, https://doi.org/10.1016/j.polymdegradstab.2024.110709*

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization